

Ajeenckya Mahadik

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Skills

Languages & Backend: Python, SQL, FastAPI, Streamlit, Git.

ML & Deep Learning: Scikit-learn, PyTorch, TensorFlow, XGBoost, Bayesian Modeling, Model Evaluation, Fine-Tuning.

LLM & AI Systems: RAG, Knowledge Graph RAG, Agentic Workflows, Transformers, LangChain, RLHF, Hugging Face.

Data, Cloud & MLOps: PostgreSQL, pgvector, AWS (S3, IAM, Lambda), Docker, Kubernetes, CI/CD (GitHub Actions).

Work Experience

University of Wisconsin–Madison

Graduate Research Assistant

Madison, WI

2025 – Present

- Built an end-to-end machine learning forecasting pipeline integrating weather signals, news derived features, feature engineering, model evaluation, and **FastAPI model** serving, improving accuracy and calibration by **12%**.
- Formulated a **Bayesian calibration** layer over a policy network forecasting model, achieving a **0.176 Brier score**, improving recall to **84%**, and reducing scenario variance by **17%**.
- Trained and evaluated **XGBoost** weather-impact models across temperature, humidity, and precipitation features, enabling production-ready inference with more stable predictions.

Fiat India Automobiles Pvt. Ltd.

Graduate Associate Engineer

India

2023 – 2024

- Developed & evaluated **Scikit-learn regression models** to predict & identify delay prone automobile operations, enabling early workload balancing and improving operational efficiency by **9%**.
- Implemented ML ready training datasets from noisy operational logs using schema standardization, missing value imputation, and **feature engineering** across delay, throughput, variance, and risk.
- Structured **AWS data processing workflows** using S3, Lambda, IAM, and CloudWatch to validate inputs, transform parameters, and generate decision support outputs at scale.

Powertrac

Analytics Engineer

India

2022 – 2023

- Engineered **predictive maintenance pipelines** from raw diagnostic logs and service records, extracting failure frequency, downtime duration, and fault recurrence features to reduce unplanned downtime by **12%**
- Developed a **risk-score driven job prioritization model**, replacing manual triage with data-informed routing and improving technician efficiency across service operations
- Established a model monitoring loop to identify low-confidence predictions, trace root causes to upstream feature logic, and drive iterative improvements in production performance

Tata Motors

Graduate Engineer

India

2021 – 2022

- Designed **ML feature pipelines** over high-volume operations data, standardizing noisy timestamps, throughput logs, and performance metrics into structured training datasets for downstream modeling
- Deployed **anomaly detection workflows** to proactively surface throughput deviations and launch readiness failures, improving system efficiency by **18%** and throughput by **15%**
- Built a data-driven process optimization framework using historical pattern modeling and impact simulation, reducing operational costs by **20%** and earning the **Kaizen Award**

Projects

Autonomous CLI Coding Agent

- Engineered a **framework-free LLM coding agent** using raw APIs with a multi-step tool calling loop across **6 workspace tools** and safety guardrails, **40%** faster than LangChain-based alternatives.

Self-Improving LLM Agent

- Integrated a self improving long-horizon **LLM agent** across AgentBench and WebArena, using trace logging, failure labeling, strategy-memory retrieval, GPT-4 evaluation, ReAct baselines, LLaMA3-8B inference, and **QLoRA** 4 bit quantization.

AI Document Chat

- Designed an asynchronous **RAG pipeline** using pgvector for chunking, embeddings, vector indexing, and cosine retrieval also built a **FastAPI and Streamlit** document QA app containerized with Docker and CI/CD workflows.

Education

University of Wisconsin–Madison

Master of Science Industrial Engineering, Specialization in AI/ML

Madison, WI

May 2026

Coursework: Machine Learning, AI Systems, Deep Learning, NLP, Trustworthy AI, AI Agents

Shivaji University

Bachelors of Engineering Mechanical Engineering

India

May 2020